

Alternative Risk-Free Rates and the New Indexation Framework: An Analysis of Compounded Overnight Rates with Applications to FRNs and Swaps

Pietro Sgrigna

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Abstract

This paper analyzes the transition from traditional IBOR benchmarks to alternative risk-free rates (RFRs), focusing on the new indexation framework based on compounded overnight rates. The study combines institutional background, literature review, and mathematical modeling to examine the implications of the reform for pricing and risk management. Building on the framework proposed by Lyashenko and Mercurio, the paper adopts a generalized Forward Market Model completed within an HJM structure, allowing a coherent and arbitrage-free representation of backward-looking rates, bond prices, and the full term structure. The theoretical framework is applied to the valuation of an RFR-linked floating rate note and a plain vanilla payer swap. A numerical simulation based on an OIS curve shows that the floating rate note remains close to par with limited sensitivity to curve shifts, while the swap exhibits a larger exposure to interest-rate movements. Although the analysis is based on a simplified parametric specification, the results confirm that the proposed framework provides a consistent and tractable approach to pricing and risk analysis in the post-IBOR environment.

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1 Introduction

The interest rate benchmark reform is one of the most relevant changes of recent years, in fixed income markets. For decades, rates such as LIBOR (London Inter-Bank Offered Rate) and EURIBOR have been the key reference in the determination of cash flows of a wide range of financial instruments, both cash and derivatives. However, after the global financial crisis and the scandals related to the benchmark manipulation, the need for a transition away from these rates emerged. The aim was to develop new indicators that would be more transparent, robust and linked to real transactions. While EURIBOR was reformed and continues to be published by the European Money Market Institute as the benchmark rate for unsecured transactions, LIBOR has been discontinued in 2023.

In this context, Alternative risk-free rates, have been introduced, such as SOFR (Secured Overnight Financing Rate, in the United States), SONIA (Sterling Overnight Index Average, in the United Kingdom) and €STR (Euro Short-Term Rate, in the European Union). These new rates are all overnight benchmarks grounded in observed overnight transactions^{1,2,3}, while the unsecured rates such as LIBOR were term rates, based on panel bank submissions regarding unsecured funding conditions.

This difference implies that the transition from IBORs to Alt-RFRs cannot be interpreted just as a simple index substitution. It entails a change in the benchmark nature, in the timing of the rate determination and in the method in which cash flows are determined. Specifically, while IBORs were forward-looking rates quoted for different tenors, the new overnight benchmark rates often require the construction of compound rates set in arrears, based on daily fixings.

This difference affects not only the cash flow construction, but also on the analysis of instruments like floating rate notes and interest rate swaps, as well as more complex interest rate derivative contracts.

The aim of this work is to analyze the new indexation framework linked to the alternative risk-free rates from both a financial and a mathematical point of view. The paper combines the institutional background, reference literature, mathematical approach and a simple illustrative example.

The remainder of the paper is organized as follows. Section 2 presents a concise, but essential review of the relevant literature, while Section 3 is centered on the formal and mathematical framework, building on the literature discussed in Section 2 with a focus on pricing and sensitivity analysis under the new indexation. Section 4, includes an illustrative application of what has been addressed in the previous sections using a floating rate note and a simple interest rate derivative as case studies to highlight the implications of the new indexation under a pricing and risk management lens.

2 Literature review

2.1 Benchmark reform and the transition from IBORs to the Alt-RFR

Benchmark reform is generally described in the literature as a structural transformation of the rates market and not just as a simple technical substitution of a framework with another. In this perspective, new benchmarks like SOFR, SONIA and €STR do not represent only new reference indexes, but also a different way to build and interpret the relevant rates for pricing and risk management.^{4,5}

AIFIRM Position Paper emphasises the fact that the reform did not only have effects on the discontinuation of LIBOR, but more broadly impacted rates market structure, indexation conventions and cash and derivatives instruments valuation processes. This interpretation is coherent with the framework provided by the BIS, which describes the transition from IBOR to the new RFR as a real paradigm shift, due a different construction of these benchmarks compared to the old interbank rates. Therefore they cannot be considered perfect substitutes.^{4,5}

2.2 Fallback rates, term adjustment and spread adjustment

One of the essential issues in the literature about the benchmark reform involves the construction of the fallback rates. AIFIRM Position Paper defines the fallback rate as the rate that substitutes the original benchmark in the case that the latter is discontinued or loses representativeness. This rate is built through two principal components: term adjustment and spread adjustment.⁴

Spread adjustment has the function of reducing value transfers between counterparties that could derive from a direct substitution of an IBOR benchmark with an overnight one. The problem arises from the fact that old IBORs embedded a credit risk and unsecured funding component, while the new RFR do not present the same pattern. According to this perspective, quantitative literature on fallback rates shows that the different proposed solutions are not equivalent, neither from an economic nor a stability of contracts value point of view. Specifically, Henrard shows how the choice of the fallback rate influences directly cash flow profile and mark-to-market of legacy instruments and how no solution would be perfectly neutral with respect to all the desirable criteria.^{4,6}

Term adjustment responds to the problem of the benchmarks different nature. While the old IBOR were term rates quoted for specific tenors, the new Alt-RFR are overnight benchmarks based on transaction with one day maturity. Consequently, in order to be used on contracts originally witten on term rates, the new benchmarks have to be converted into an economic comparable rate. As the AIFIRM underlines, this can be done with a forward-looking approach, based on market expectation at the beginning of the period, or with a backward-looking one, based on overnight fixings actually observed and compounded in the period of interest.⁴

2.3 Forward-looking and backward-looking rates

The discrimination between forward-looking and backward-looking rates represents one of the most relevant nodes in the recent literature. Forward-looking rates are known at the beginning of the period of interest and reflect, at least in theory, the market expectation on the rate level in the following period. Backward-looking rates, on the other hand, are set in arrears based on overnight fixings quoted during the period and compounded afterwards.^{4,5} This distinction has direct implications on the contractual and evaluating level. AIFIRM Position Paper observes, as a matter of fact, that the adoption of overnight benchmark implies the need to introduce specific operational conventions, such as lookback period, payment delay of last reset, in order to make the construction of the backward-looking rate compatible with the payment and contracts administration requirements.⁴

On a theoretical level, Backwell, Macrina and Schlögl show that the distinction between term rates and overnight benchmark cannot be reduced to a simple maturity difference, but is strictly connected to the multicurve framework and to the presence of basis spread between term structures, built on different payment frequency. In this sense, the transition to overnight benchmark highlights that traditional term rates are not pure aggregates of overnight rates, while equally reflecting risk component and more complex market structure information.⁷

2.4 Lyashenko and Mercurio contribution

While institutional literature insists mainly on the operational and regulatory aspects of benchmark reform, Lyashenko and Mercurio contribution is particularly relevant for the theoretical foundation of the new backward-looking rates. In their first work, the authors show that the transition from traditional LIBOR-like term rates to the new setting-in-arrears rates does not require abandoning the classic forward-rate approach, but rather extending it in a coherent way⁸. The central idea of the paper consists in introducing an extended forward rate $R_j(t)$ which allows linking the forward-looking rate, expected at the beginning of the period, to the backward-looking rate, made at the end of the same period, with one single process. Before the start of the accrual period, this process coincides with the classic forward-looking term rate. During accrual period it progressively integrates the observed overnight fixings. At the end of the period of interest, this rate corresponds with the actually realized rate compounded in arrears. In this way, the authors prove that the new overnight rates context can be treated inside a generalization, which matches the traditional market models framework.⁸

2.5 Multicurve framework and curve construction

In order to fully comprehend the implications of the benchmark reform on instruments valuation, it is useful to recall the literature about the multicurve framework. Ametrano and Bianchetti show that, in the post-crisis context, plain vanilla derivatives pricing can no longer be performed coherently using only one curve, but requires a distinction between discounting and forwarding curves, built as function of funding and market conventions.⁹ This contribution

is particularly important for this work, since the benchmark reform is part of a framework in which the separation in discounting and forecasting already emerged as a structural element in the modern derivatives pricing. In this perspective, the transition to overnight rates not only modifies the floating benchmark, but connects to the growing importance of OIS/RFR curves as a natural basis for collateralized contracts discounting. The AIFIRM Position Paper itself provides extensive coverage about the effects of the reform on curve construction, OIS discounting and derivatives valuation, confirming the strong link between benchmark reform and multicurve framework.^{9,4}

2.6 Generalized Forward Market Model and curve completion

The later work from Lyashenko and Mercurio completes this theoretical track showing how the generalized Forward Market Model can be extended to describe the whole term structure. Precisely, the authors show how it is possible to generate in an arbitrage-free way, not only a discrete set of forward term rates, but also bond prices $P(t, T)$ and the bank account $B(t)$ and, more broadly, every point on the yield curve.¹⁰

The significance of this result, for the present work, is not merely conceptual. As a matter of fact, it confirms that the new overnight benchmarks and the compounded in arrears rates do not represent a conflicting deviation from the modern rates theory, but can be integrated in a coherent and arbitrage-free framework, suitable for connecting forward term rates, bond prices and bank account. To this end, the generalized FMMM and his HJM completion represent the baseline theoretical context both for the mathematical treatment of this work and for the following numerical analysis of instruments such as floating rate notes and interest rate derivatives.¹⁰

2.7 Present study in the literature framework

Altogether, the examined literature allows to distinguish three levels of analysis. The first one is the institutional and operational level, in which the benchmark reform is interpreted as a transition process from the old interbank benchmarks to the new and more robust overnight ones. The second one is the qualitative and contractual level, where take prominence fallback rates, term adjustment and spread adjustment. The third one is the properly modeling one, in which the new backward-looking rates are integrated in a framework coherent with the term-structure modeling.^{4,6,8,10}

The present study lies at the intersection of these three levels, but with a limited aim. It focuses on the construction of the new indexation framework linked to the Alt-RFR and on the analysis of the implications, both from a financial and mathematical point of view. The paper makes specific reference to a floating rate note and a plain vanilla interest rate swap, which are the products used also in the simulation. In this perspective, the generalized FMM and its HJM completion are not just reminded as a theoretical background, but serve as the reference framework for the mathematical discussion and the following numerical simulation.

3 Mathematical treatment

3.1 Overnight rate compounded in arrears

The starting point of the mathematical discussion is the overnight rate compounded in arrears on the period of interest $[T_{j-1}, T_j]$. In the new benchmark reform framework, many contracts written on RFR are based on ex-post built rates computed on the overnight fixings observed during the period, instead of term-rates known at its beginning.

Stating with τ_j the year fraction for the period $[T_{j-1}, T_j]$, the compounded rate can be expressed, in a discrete form, as:

$$R(T_{j-1}, T_j) = \frac{1}{\tau_j} \left(\prod_{i=1}^n (1 + r_i \delta_i) - 1 \right),$$

where r_i represents the overnight fixing relative to the daily subrange i and δ_i is the corresponding day-count fraction.

In Lyashenko and Mercurio's framework this construction is approximated in continuous time as

$$R(T_{j-1}, T_j) = \frac{1}{\tau_j} \left(e^{\int_{T_{j-1}}^{T_j} r(u) du} - 1 \right) = \frac{1}{\tau_j} \left(\frac{B(T_j)}{B(T_{j-1})} - 1 \right),$$

where $r(u)$ is the instantaneous risk-free rate and $B(t)$ is the bank account. This expression is particularly useful, since it allows to directly link the setting-in-arrears compounded rate to the classical rates term-structure.

3.2 Forward-looking and backward-looking rates

The distinction between forward-looking and backward-looking rates is pivotal in the new indexation framework. The backward-looking rate $R(T_{j-1}, T_j)$ is the realized rate on the period $[T_{j-1}, T_j]$ and become known only at the end of the period. The corresponding forward-looking rate can be defined, instead, as the rate known at the beginning of the period, which results economically equivalent, thanks to the absence of arbitrage, at the realized ex-post rate.

Formally, Lyashenko and Mercurio define the forward-looking rate as

$$F(T_{j-1}, T_j) = \mathbb{E}^{T_j} [R(T_{j-1}, T_j) \mid \mathcal{F}_{T_{j-1}}],$$

i.e. as the expectation, under T_j -forward measure, of the backward-looking realized on the period. Conceptually speaking the forward-looking rate represents the risk-neutral forecast of the rate compounded in arrears that is going to be observed at the end of the period.

This relationship is fundamental, since it clarifies that forward-looking and backward-looking rates should not be interpreted as disconnected entities, but rather as two different temporal perspectives for describing the same period rate: the former ex-ante, the latter ex-post.

3.3 Extended forward rate

The core theoretical contribution of the first paper from Lyashenko and Mercurio consists of the introduction of a unique process $R_j(t)$ able to link in a coherent way the forward-looking and the backward-looking rates.

This process is defined as

$$R_j(t) = \frac{1}{\tau_j} \left(\frac{P(t, T_{j-1})}{P(t, T_j)} - 1 \right),$$

where $P(t, T_j)$ is the extended zero-coupon bond price with maturity T_j . This setting presents some fundamental properties:

- $R_j(t)$ is a martingale under T_j -forward measure;
- at time T_{j-1} coincides with the forward-looking fixing of the period:

$$R_j(T_{j-1}) = F(T_{j-1}, T_j);$$

- at time T_j coincides with the backward-looking realized rate:

$$R_j(T_j) = R(T_{j-1}, T_j);$$

- for $t \geq T_j$ stays constant, since the final fixing is already known.

When t is inside the accrual period, so for $T_{j-1} < t < T_j$, the rate $R_j(t)$ integrates simultaneously the part of the overnight benchmark which is already realized and the still expected part for the residual period. In this regard, the extended process gives a natural representation of the continuous transition between the forward-looking term rates framework and the backward-looking compounded rates one.

3.4 The generalized Forward Market Model dynamic

Once the process $R_j(t)$ is defined, the next step consist in the specification of the stochastic dynamic. In the generalized Forward Market Model, Lyashenko and Mercurio postulates the dynamic of extended forward term rates under the risk-neutral measure Q . In scalar notation, for every j , one can write

$$dR_j(t) = \sigma_j(t)\gamma_j(t) \sum_{i=\eta(t)}^j \rho_{i,j} \frac{\tau_i \sigma_i(t) \gamma_i(t)}{1 + \tau_i R_i(t)} dt + \sigma_j(t)\gamma_j(t) dW_j(t),$$

where $\sigma_j(t)$ represents the volatility of $R_j(t)$, $\rho_{i,j}$ the instantaneous correlation between different forward term rates and $\gamma_j(t)$ is a deterministic function of volatility decay. The $\gamma_j(t)$ has a fundamental role in the model:

- $\gamma_j(t) = 1$ before the start of the period $[T_{j-1}, T_j]$;

- $\gamma_j(t)$ decreases during the accrual period;
- $\gamma_j(t) = 0$ for $t \geq T_j$.

In this way rate's volatility decreases progressively during the period of application, mirroring the fact that an increasing part of the final fixing has already been incorporated through the observed overnight rates. This is one of the most important differences with the classic LMM, in which the forward rate is fixed at the beginning of the period and does not evolve within it.

A notably easy specification, used by the same authors is the one with linear decay:

$$\gamma_j(t) = \min \left\{ \frac{(T_j - t)^+}{T_j - T_{j-1}}, 1 \right\}.$$

Such a choice allows to capture in an intuitive way, the idea that the extended rate is fully forward-looking before the period, then progressively less volatile during the accrual period and in the end completely fixed at T_j .

3.5 FMM-HJM completion of the model

Generalized FMM, in the depicted form, model coherently a discrete set of extended forward term rates $R_j(t)$. However, for many pricing and simulation applications this is still not sufficient. In particular, to price instruments indexed on compounded overnight benchmarks it is necessary to be able to describe not only a finite set of rates, but also bond prices $P(t, T)$, the bank account $B(t)$ and, more broadly, the full term structure of rates.

For this reason, in their second paper on this subject, Lyashenko and Mercurio show how the generalized FMM could be integrated into a markovian HJM framework, obtaining a complete and arbitrage-free term structure model. The idea at the base is that the discrete process $R_j(t)$, defined over periods $[T_{j-1}, T_j]$, should not be considered as isolated objects, but as components of a broader structure, which can generate the bond prices and the whole curve dynamic.

The starting point of such completion is the relation, already introduced in the first paper, between the extended forward rate and the bond prices

$$R_j(t) = \frac{1}{\tau_j} \left(\frac{P(t, T_{j-1})}{P(t, T_j)} - 1 \right).$$

This expression is crucial since connects directly the discrete family of extended term rates to the structure of zero-coupon bonds. Specifically, it implies the inverse relation

$$P(t, T_j) = \frac{P(t, T_{j-1})}{1 + \tau_j R_j(t)},$$

that allows to reconstruct recursively bond prices on the grid dates, once that the processes $R_j(t)$ are known. However, this reconstruction is still not sufficient to determine the whole curve $P(t, T)$ for arbitrary maturities, nor to generate the short rate and the bank account in a complete way.

In the complete framework a single-curve context is assumed, in which the instantaneous risk-free rate $r(t)$ determines the money-market account according to

$$dB(t) = r(t)B(t) dt, \quad B(t) = \exp \left(\int_0^t r(u) du \right).$$

Define also, for every maturity T , the extended bond price as

$$P(t, T) = \exp \left(- \int_t^T f(t, u) du \right),$$

where $f(t, T)$ is the instantaneous forward rate and holds

$$f(t, u) = r(u), \quad t \geq u.$$

In this way, the extended bond stays defined after its maturity and coincides with the value of the self-financing strategy that reinvests the unitary payoff in the bank account. Specifically for $T = 0$ one has

$$P(t, 0) = B(t).$$

At this point, the second paper introduces an extended Markovian HJM. In the classic HJM formulation, for every maturity T the instantaneous forward rate evolves under the risk-neutral Q measure following

$$df(t, T) = \sigma(t, T)^\top \int_t^T \sigma(t, s) ds dt + \sigma(t, T)^\top dW(t).$$

To make this structure coherent with extended bonds, Lyashenko and Mercurio expand the dynamic to the time steps following the maturity, setting the volatility equal to zero for $t \geq T$:

$$df(t, T) = \mathbf{1}_{\{t \leq T\}} \left[\sigma(t, T)^\top \int_t^T \sigma(t, s) ds dt + \sigma(t, T)^\top dW(t) \right].$$

The indicator function $\mathbf{1}_{\{t \leq T\}}$ has the effect of making the process constant after T , such that

$$f(t, T) = r(T), \quad t \geq T.$$

as already seen before. Applying Ito's lemma and Fubini theorem, the authors get the risk-neutral dynamic of extended bonds:

$$\frac{dP(t, T)}{P(t, T)} = r(t) dt - \left(\int_t^T \sigma(t, u) \mathbf{1}_{\{t \leq u\}} du \right)^\top dW(t).$$

Specifically, for $t > T$, this expression reduces to

$$\frac{dP(t, T)}{P(t, T)} = r(t) dt,$$

coherently with the extended bond interpretation of a self-financing strategy that, once the maturity is exceeded, evolves like a bank account.

As seen in the previous section, generalized FMM yields, under Q , the extended term rates dynamics:

$$dR_j(t) = \gamma_j(t) \sigma_j^R(t)^\top \sum_{i=\eta(t)}^j \sigma_i^R(t) \frac{\tau_i \gamma_i(t)}{1 + \tau_i R_i(t)} dt + \gamma_j(t) \sigma_j^R(t)^\top dW(t),$$

where $\sigma_j^R(t)$ is the vector of the rate's $R_j(t)$ volatilities, $W(t)$ is a multi-dimensional Brownian and $\gamma_j(t)$ is the volatility decay function. This structure allows the rate to continue evolving in a stochastic way inside the accrual period and to fix only at time T_j .

The problem at this point is to join the discrete dynamics of $R_j(t)$ with continuous dynamic of the instantaneous forward rates $f(t, T)$. To achieve that, the second paper assumes a volatility characterized by a separable form:

$$\sigma(t, T) = \sum_{k=1}^M \varsigma_k(t) g_k(T),$$

or in a form by component

$$\sigma_k(t, T) = \varsigma_k(t) g_k(T).$$

This choice is crucial because makes the model markovian: the dynamic of the curve can, in fact, be expressed as a function of a finite number of state factors. Specifically, the authors show that the integrated dynamic of instantaneous forward rates can be written as

$$f(t, T) = \begin{cases} f(0, T) + g(T)^\top X(t) + g(T)^\top Y(t) G(t, T), & t < T, \\ r(T) = f(0, T) + g(T)^\top X(T), & t \geq T, \end{cases}$$

where the state processes $X(t)$ and $Y(t)$ satisfy

$$dX(t) = Y(t) g(t) dt + \varsigma(t) dW(t), \quad dY(t) = \varsigma(t) \varsigma(t)^\top dt,$$

with initial condition

$$X(0) = 0, \quad Y(0) = 0.$$

Here:

- $g(T)$ is the vector of functions $g_k(T)$;
- $G(t, T)$ is the vector of the corresponding integrated functions;
- $X(t)$ is the vector of markovian factors;
- $Y(t)$ is the matrix of quadratic terms associated to the volatility.

Substituting this representation in the bond prices formula, the authors obtain

$$P(t, T) = \begin{cases} P(0, t, T) \exp\left(-G(t, T)^\top X(t) - \frac{1}{2}G(t, T)^\top Y(t)G(t, T)\right), & t < T, \\ \exp\left(\int_T^t r(u) du\right), & t \geq T, \end{cases}$$

where

$$P(s, t, T) = \frac{P(s, T)}{P(s, t)}$$

is the forward discount factor observed at time s between t and T .

This is the real completion of the model. From extended term rates $R_j(t)$, generalized FMM is merged with a HJM markovian structure with separable volatility that allows to describe coherently:

- extended forward term rates $R_j(t)$;
- zero-coupon bond prices $P(t, T)$ for all relevant maturities;
- the continuously accruing money-market account $B(t)$;
- and, therefore, the evolution of the whole rates term structure.

The importance of this completion is especially evident for instruments like floating rate notes and interest rate swaps indexed to compounded overnight benchmarks. The coherent pricing of these contracts requires, in fact, not only the simulation of fixings on single periods but also the availability of discount factors along every simulated path. In this sense the FMM-HJM completion does not represent only theoretical background, but also the structural reference to link rates dynamic, bond prices and bank account in a single model.

This framework will be used in this work as the theoretical base for the following numerical simulation. The main idea is the one proposed by Lyashenko and Mercurio in the two papers analyzed: starting from a discrete set of extended term rates, completing it with an arbitrage-free term structure and using the resulting model to price a floating rate note and a plain vanilla interest rate swap.

3.6 Valuation of an RFR-linked floating rate note

Let us consider a floating rate note with nominal N , payment dates $T_1, \dots, T_n$ and coupon indexed to the overnight benchmark compounded on each accrual period, increased by a constant spread s . The coupon paid at time T_j can be written as

$$\text{Coupon}_j = N\tau_j(R_j(T_j) + s),$$

where $R_j(T_j)$ coincides with the backward-looking rate realized over the period $[T_{j-1}, T_j]$.

In the framework developed above, such rate is not exogenous but is generated by the same risk-free arbitrage-free model that determines bond prices, bank account and extended term

rates dynamic. In this sense, the valuation of the floating rate note lies in the unitary context obtained by HJM-FMM completion, where floating benchmark and discount factors derive from the same risk-free term structure.

Indicating with:

$$\eta(t) = \min\{j : T_j \geq t\}$$

the index of the first payment date not preceeding t , the FRN value at time t is given by the present value of residual future cashflows, plus the redemption of notional at maturity:

$$V_{\text{FRN}}(t) = N \sum_{j=\eta(t)}^n \tau_j P(t, T_j) (R_j(t) + s) + NP(t, T_n).$$

This expression represents a direct application of the discounted cash flows principle to an instrument indexed to compound overnight benchmark. From an economic standpoint, it highlights how, in the new indexation framework the floating coupon is not linked anymore to a IBOR-like forward-looking rate, but is connected to a rate determined in arrears by capitalization of the overnight benchmark over the accrual period.

3.7 Valuation of an RFR-linked fixed-floating interest rate swap

Let us consider now a plain vanilla swap in which the floated leg pays, at every date T_j , the overnight rate compounded over the period $[T_{j-1}, T_j]$, while the fixed leg pays a constant rate K over a possibly different calendar. In the framework developed above, this floating rate is represented by the corresponding extended term rate $R_j(t)$, defined by

$$R_j(t) = \frac{1}{\tau_j} \left(\frac{P(t, T_{j-1})}{P(t, T_j)} - 1 \right).$$

As already seen for the FRN, also the swap pricing lies in the same framework obtained through HJM-FMM completion.

Indicating with $T_{a+1}, \dots, T_b$ floating leg payment dates and with $T'_{c+1}, \dots, T'_d$ fixed leg payment dates, for $t < T_{a+1}$ the value of the swap for the fixed rate payer is given by

$$V_{\text{swap}}(t) = \sum_{j=a+1}^b \tau_j P(t, T_j) R_j(t) - K \sum_{j=c+1}^d \tau'_j P(t, T'_j),$$

where τ_j and τ'_j indicates the year fractions of the two legs.

Imposing $V_{\text{swap}}(t) = 0$, one gets the par swap rate:

$$S(t) = \frac{\sum_{j=a+1}^b \tau_j P(t, T_j) R_j(t)}{\sum_{j=c+1}^d \tau'_j P(t, T'_j)}.$$

Therefore, the swap formula confirms the pricing coherence with the unitary framework ad-

dressed before, also in the case of derivatives.

3.8 Sensitivity measures

To complete the mathematical framework, it is useful to introduce some simple sensitivity measures, which will be applied in the next section for the numerical simulation. In this work, such sensitivities will be computed with a finite difference approach (or bump-and-reprice), coherent with the model structure and its following numerical implementation.

Since, in the adapted framework, both bond prices and extended term rates depend on the same risk-free term structure, sensitivities will mainly refer to the movements of that curve.

DV01 / PV01 The most immediate measure is the DV01 (or PV01), i.e. the variation of the value of the instrument in response to a parallel shift of 1 basis point of the relevant curve. Indicating with $V(\mathbf{z})$ the value of the instrument as a function of the initial zero rates vector $\mathbf{z}$, and with $\mathbf{1}$ the unitary vector, let us define

$$\text{DV01}(t) = \frac{\partial V(t; \mathbf{z})}{\partial z^{\parallel}} 10^{-4},$$

where $z^{\parallel}$ represents a parallel shift of the curve. Numerically, its finite differences approximation is

$$\text{DV01}(t) \approx \frac{V(t; \mathbf{z} - 10^{-4}\mathbf{1}) - V(t; \mathbf{z} + 10^{-4}\mathbf{1})}{2}.$$

This measure gives the overall sensitivity of the price with respect to small uniform movements of the curve. DV01 can be identified as the parallel delta sensitivity to the discounting curve, while for IRS, there is also the parallel delta sensitivity to the forward curve, often defined as PV01. In this work, this distinction reduces to just one sensitivity, since since a risk-free single-curve context is considered.

Convexity At last, convexity can be introduced. It represents a second order simple measure with respect to a parallel shift of the curve, useful as a numeric control on the quality of the linear approximation which is implicit in DV01. Defining convexity with respect to a parallel shift as:

$$\Gamma(t) = \frac{\partial^2 V(t; \mathbf{z})}{\partial (z^{\parallel})^2},$$

its finite differences approximation is

$$\Gamma(t) \approx \frac{V(t; \mathbf{z} + 10^{-4}\mathbf{1}) - 2V(t; \mathbf{z}) + V(t; \mathbf{z} - 10^{-4}\mathbf{1})}{(10^{-4})^2}.$$

Consequently, for a small parallel shift Δz , the price variation can be approximated to the second order by

$$\Delta V(t) \approx -\text{DV01}(t) \frac{\Delta z}{10^{-4}} + \frac{1}{2}\Gamma(t)(\Delta z)^2.$$

In the context of products analyzed in this study, such measure does not constitute the main risk factor, but can be useful to verify the potential presence of a modest non-linearity due to discounting.

Overall, these risk measures allow to directly link between theoretical indexation framework and risk management, showing how risk-free initial curve variations can spill over the price evaluation of a floating rate note and a swap, both indexed to compound overnight benchmarks.

4 Numerical Implementation and Simulation

The proposed numerical implementation aims to operationally transpose the HJM-FMM framework discussed in the previous sections, in order to evaluate financial instruments indexed to risk-free rates, in a context consistent with the new post-IBOR paradigm. The procedure is articulated in three steps: construction of the initial OIS curve, extended rates $R_j(t)$ dynamic simulation and pricing with analysis of the main risk measures.

The starting point is the discount factor curve, built from OIS data, fixed at October 2019. From it, the zero rates are extracted first and then, for every interval $[T_{j-1}, T_j]$, the initial extended rates $R_j(0)$, which represent the continuous approximation of the set-in-arrears rates. This choice allows us to treat instruments which are based on overnight RFR within the same dynamic structure as traditional forward market models, while remaining consistent with the backward-looking nature of the new benchmarks.

The dynamics are implemented in a hybrid form. From the HJM side, the evolution of the instantaneous curve is driven by a set of Gaussian factors $X_k(t)$, which allow us to describe in an arbitrage-free way the dynamics of instantaneous forward rate. From the FMM perspective, instead, these factors are transferred to the extended rates $R_j(t)$, which constitute the relevant quantities for the pricing of the compounded set-in-arrears flows. In numerical terms, the model is discretized on a monthly grid and, at every step, state factors $X(t)$, short rate $r(t)$, bank account $B(t)$ and $R_j(t)$ are updated on the different maturities.

A central aspect of the implementation concerns the specification of the volatility. In line with the framework of Lyashenko and Mercurio, the extended rates volatility is assumed to depend both on the initial level of the rate and on the residual maturity of the relative accrual period. Specifically, this study adopts a parametric form

$$\sigma_j(t) = (a_0 + a_1 R_j(0)) \gamma_j(t),$$

where a_0 represents a base volatility component, a_1 measures the sensitivity to the rate level, while $\gamma_j(t)$ is a temporal decay function. The latter reflects the economic idea that the uncertainty surrounding the compounded rate decreases progressively as the accrual period is realized, until it becomes small close to the final fixing date. Operatively, $\gamma_j(t)$ is implemented as a linear function of the residual time, in order to obtain a simple volatility profile which is still coherent with the problem structure.

The parameters used in the simulation ($a_0 = 0.001$, $a_1 = 0.10$), are not obtained from a calibration to market-implied volatilities, but are chosen to generate regular trajectories and economically reasonable sensitivities. In particular, a_0 sets a minimum level of volatility even for very low initial rates, while a_1 allows the volatility to rise as the rate level increases. Hence, it is an illustrative parametrization: the purpose is to verify the internal consistency of the framework and its ability to produce realistic values and risk measures and not to replicate the market structure.

On this basis, two vanilla instruments are considered: a floating rate note with a five-year

maturity, nominal amount of €100 and contractual spread of 25 basis points and a payer swap with the same maturity, initially quoted with a par fixed rate.

The results on a single simulated path are consistent with the economic intuition. The FRN has an initial value above par, approximately €101.27, thanks to the effect of the positive spread, and converges gradually to €100 as maturity approaches. The payer swap, instead, has a zero initial value, as expected for a contract entered into at a par rate, while its value along the path depends on the specific realized trajectory of the rates.

To avoid tying down the conclusion to just one scenario, the pricing has been extended to a Monte Carlo multi-path simulation. Mean values confirm that the FRN maintains a very stable profile, with negligible standard errors and a price that approaches par. The swap, on the other hand, shows more dependence on the scenario, with a higher dispersion of future values. This step clearly separates the behaviour of a single path from the average model implications.

The risk analysis is conducted using a bump-and-reprice procedure, consistent with the simulated framework. In response to a parallel shock of 1 basis point on the initial curve, discount factors, initial extended rates, and following rates dynamic are reconstructed. The results show that the FRN has a very low parallel DV01, consistent with its low duration and its periodic coupon reset. The swap, instead, shows a sensitivity with a significantly higher order of magnitude, which reduces almost linearly over time, thanks to the progressive decline of the residual swap annuity. The convexity results complement the first-order risk analysis by showing that second-order effects remain limited for the FRN, while they are substantially more pronounced for the payer swap. Within the sign convention adopted in the numerical implementation, FRN convexity is positive and small, whereas swap convexity is negative and much larger in absolute value. For both instruments, however, the absolute magnitude of convexity declines over time as residual maturity shortens.

Table 1: Monte Carlo mean values, parallel DV01, and convexity of the two instruments

t	FRN mean	Payer swap mean	FRN DV01	Swap DV01	FRN convexity	Swap convexity
0	101.2665	0.0000	0.000349	-0.050480	12.252	-2533.707
1	101.0106	0.0639	0.000228	-0.040273	6.466	-1615.827
2	100.7555	0.1590	0.000132	-0.030111	2.872	-905.390
3	100.5023	0.1995	0.000063	-0.020023	0.943	-401.053
4	100.2505	0.1633	0.000019	-0.009994	0.157	-100.000

Table 1 summarizes the main numerical results. The FRN remains close to par throughout the simulation horizon and exhibits only a negligible sensitivity to parallel curve shifts, whereas the payer swap shows a substantially larger first-order exposure to interest-rate movements.

Overall, the simulation confirms that the HJM-FMM framework can coherently represent the pricing and risk of instruments linked to alternative risk-free benchmarks. However, this study also has some limitations. First of all, the volatility is assigned parametrically and not calibrated to interest rate options. Second, the analysis only covers vanilla instruments and simplified compounding conventions. Hence, a natural extension consists in introducing an im-

plied volatility calibration, a more flexible volatility structure, and more complex instruments. Another possible extension is the use of a finer time grid, which would reduce discretization error and allow for a more accurate approximation of both the continuous-time dynamics of the model and the overnight compounding mechanism.

Nevertheless, the exercise shows how the transition to overnight benchmarks can be treated in a uniform manner inside a dynamic arbitrage-free framework.

5 Conclusions

The study showed how the reference rates reform does not imply just an operational substitution of the IBORs with new benchmarks, but also requires an adjustment of the modelling framework used for pricing and risk management. Specifically, the transition to compounded in arrears overnight rates makes the direct application of the traditional market models built on forward term rates, defined ex ante, less natural and necessitates the introduction of state variables coherent with the backward-looking nature of the new RFR.

In this context, the considered HJM-FMM framework represents a theoretically solid and numerically tractable solution. The approach allows, in fact, to model in an arbitrage-free way the dynamics of extended rates connected to the risk-free benchmark, keeping, at the same time, an efficient structure, flexible enough to allow the evaluation of interest rate instruments. The obtained numerical results confirm this coherence: the FRN shows values close to par, while the swap maintains a higher exposure to the curve movements, as economically expected.

The analysis also highlights that the transition from a single simulated path to a Monte Carlo valuation is crucial to distinguish between scenario-specific results and robust properties of the model. Despite the limits of a parametrization not calibrated by implied volatilities, the exercise shows that the adopted framework represents a convincing basis for future extensions, as discussed in the previous section.

To summarize, the new post-IBOR context requires models able to clearly incorporate the overnight rates and the linked compounded cash flows structures. The analyzed framework responds to this need and shows how the transition to the alternative RFR can be treated in a coherent way, not only at the institutional and contractual level but also at the quantitative one.

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